ORVL-027

Video Topology

Axiom Video Topology · Constitutional Video Intelligence · Temporal Object-Event Representation

> *"Video AI asks what pixels are in this frame. Axiom asks what object is this, where is it, how did its surface move, what changed, and what event did that create — then stores only the answer."*

Concept Note · Antonio Roberts · Orivael · June 2026

Status: CONCEPT — Software implemented (axiom_video/)	**Pairs with:** ORVL-025 (Event Token) · ORVL-026 (Audio Groove)	**Domain:** Video · Robotics · Physical-World Learning

Abstract.

Axiom Video Topology converts a frame sequence into a compact temporal topology — tracked object identities, surface/pose states, motion trajectories, depth relations, impact events, and a cause-and-effect event chain. Specialist micro-agents each analyse one aspect of the visual signal; a coordinator merges them into a signed `VIDEO_TOPOLOGY_BLOCK`. The block is inspectable, routable, verifiable, and evidence-backed — it stores the event meaning without dragging every frame forward. The existing implementation in `axiom_video/` ships: SceneGraph ingestion (PIL / NumPy / pure-Python frames), ObjectTracker (IoU-based cross-frame identity), MotionClassifier (velocity + direction), ColorWatcher, DepthEstimator, ImpactDetector (deceleration + contact IoU), SurfaceClassifier, and TemporalChainExtractor — all producing HMAC-signed reports.

01 | VIDEO AS TEMPORAL TOPOLOGY

1. Video as Temporal Topology

A UV map connects surface points to texture coordinates. A video topology map connects objects, surfaces, screen positions, motion paths, timing, and meaning — the same relationship, extended through time. The system stores not just what pixels appear in a frame, but what object they belong to, how that object moved, and what event its motion caused.

Video Element	Topology-style Feature	AI Meaning
Frame sequence	Time-indexed visual states	Before, during, after
Object region	Tracked entity ID	Cup, hand, door, tool
Surface / pose	Shape and orientation map	Tilt, rotation, contact

Video Element	Topology-style Feature	AI Meaning
Motion path	Trajectory over time	Reach, lift, fall, collide
Depth / scale	Spatial relation estimate	Near, far, behind, above
State change	Object condition shift	Closed to open, upright to tilted
Event chain	Cause-and-effect graph	Hand grips cup, cup tilts, liquid spills

02 | VIDEO MICRO-AGENT STACK

2. Video Micro-Agent Stack

Video is divided into specialist agents. Some agents read keyframes, some track objects, some estimate surface orientation, some detect motion, and others infer events or cause-and-effect. The coordinator combines those reports into a compact `VIDEO_TOPOLOGY_BLOCK`.

Agent	Tracks	Example Output
Frame Agent	Keyframes, scene cuts, visual stability	Important frames at 0s, 1.2s, 2.8s
Object Agent	Objects and identities across frames	hand_01 and cup_01 detected
Surface Agent	Pose, orientation, contact points	cup rotates from upright to tilted
Motion Agent	Velocity, trajectory, direction	hand approaches cup from left
Depth Agent	Near / far relation and occlusion	hand is in front of cup
Event Agent	Action summaries from motion + contact	hand grips cup
Causality Agent	Likely cause-and-effect chain	tilt caused liquid spill

> **Topology Map**
> The coordinator's Topology Map is not a frame buffer — it is an ordered event graph: objects x time x meaning. A downstream agent reading cup_01: upright_to_tilted -> event: spill needs zero pixels to reason about what happened and why.

03 | VIDEO_TOPOLOGY_BLOCK FORMAT

3. VIDEO_TOPOLOGY_BLOCK Format

Each video analysis result is serialised as a compact block. The block carries tracked objects, surface states, the motion chain, a top-level event with confidence, an evidence trace listing contributing agents, and an HMAC-SHA256 signature. Signature failure refuses the block at any inspection point.

```
VIDEO_TOPOLOGY_BLOCK
source: clip_001
duration: 4.8s
keyframes: [0, 12, 24, 39, 61, 88]
objects: [hand_01, cup_01, liquid_01]
surface_state: { cup_01: upright_to_tilted }
motion_chain: hand_01 approaches -> grips -> cup_01 tilts
event: { action: spill, cause: cup_tilt, confidence: 0.86 }
evidence_trace: [object_agent, surface_agent, motion_agent, causality_agent]
signature: <HMAC-SHA256 over canonical JSON, axiom-video-temporal-v1>
```

Field	Content
keyframes	Frame indices of scene cuts and visually significant moments
objects	Stable cross-frame identity labels assigned by ObjectTracker
surface_state	Per-object orientation change (e.g. upright_to_tilted, open_to_closed)
motion_chain	Ordered action sequence connecting object trajectories
event	Top-level inferred event: action label, causal object, confidence
evidence_trace	Ordered list of contributing agent names — the audit chain
signature	HMAC-SHA256 under axiom-video-temporal-v1; domain-isolated from audio key

04 | IMPLEMENTED AGENTS (AXIOM_VIDEO/)

4. Implemented Agents (axiom_video/)

The `axiom_video/` package ships the full seven-agent pipeline. Each module produces a signed report; the `TemporalChainExtractor` is the coordinator that merges tracks + motions + impacts into the final event chain.

Module	Agent Role	Key Output
ingest.py	Frame Agent	SceneGraph from PIL / NumPy / pure-Python frames; duck-typed, no hard deps
object_tracker.py	Object Agent	ObjectTrackReport: stable cross-frame Track IDs via IoU greedy matching (threshold 0.3) + label equality; HMAC-signed under axiom-video-objects-v1
motion.py	Motion Agent	MotionReport: per-track velocity vectors, motion class (approaching / receding / lateral / static / fast)
surface.py + depth.py	Surface + Depth Agents	Surface orientation and depth / occlusion relations between tracked objects
color_watcher.py	Frame Agent (color)	Per-object dominant-color shift over time — detects material state changes
impact.py	Event Agent	ImpactReport: deceleration events (velocity drop) and contact events (IoU overlap + velocity history)
temporal_chain.py	Causality Agent + Coordinator	TemporalChainReport: ordered TemporalEvent list (appear / motion_start / contact / motion_change) signed under axiom-video-temporal-v1

> **Physical-world learning example**
> A child's toy knock-over: hand_01 approaches contact with cup_01 (IoU overlap + velocity drop) cup_01 surface_state: upright_to_tilted event: spill, cause: cup_tilt, confidence: 0.86. The full causal chain is captured in under 200 bytes of structured JSON without storing a single pixel.

05 | CROSS-PATENT WIRING

5. Cross-Patent Wiring

Patent	Role for Video Topology
ORVL-004 MKB	Each micro-agent report registers as a KnowledgeBlock (`block_type=VIDEO_TOPOLOGY`). The TemporalChainExtractor's merged block is the composed output certified by CBV before downstream use.
ORVL-010 CBV	Topology thresholds (`IOU_THRESHOLD=0.3`, contact detection parameters) are CANNOT_MUTATE in each module. CBV certifies the constraint set before the agent enters the certified registry.
ORVL-014 CWM	TemporalChainExtractor's event chain is an ORVL-014 causal graph in the physical domain: cup_tilt spill is the same propagation structure as auth_block transaction_block, expressed in physical objects instead of financial blocks.
ORVL-022 CPI	ImpactDetector's contact events (deceleration + IoU overlap) directly feed the CPI PhysicalMonotonicGate: a falling-velocity tick becomes a StabilityFrame, and a sharp deceleration fires an L2-L3 reflex.
ORVL-025 Event Token	`VIDEO_TOPOLOGY_BLOCK` is the Video layer of an EventToken. objects, surface_state, motion_chain, and event map directly to the EventToken Video payload schema.
ORVL-026 Audio Groove	Audio groove (rhythm, texture, intent) and video topology (objects, causality) merge at the coordinator level into a single multimodal evidence graph — audio gives timing and tone, video gives physical cause.

6. Provisional Patent Claims (ORVL-027)

Claim 1. A method of representing a video clip as temporal topology wherein a sequence of frames is analysed to produce a VIDEO_TOPOLOGY_BLOCK comprising: stable cross-frame object identities (ObjectTracker), surface-pose state transitions per object (SurfaceAgent), an ordered motion chain connecting trajectories (MotionAgent), a top-level cause-and-effect event with confidence (CausalityAgent), an evidence_trace listing contributing agents, and an HMAC-SHA256 signature over the canonical JSON form — such that downstream agents can inspect, route, and verify the block without decoding individual frames.

Claim 2. A modular micro-agent pipeline for video analysis wherein each of seven specialist agents — Frame, Object, Surface, Motion, Depth, Event, and Causality — independently processes one aspect of the visual signal and produces a compact signed sub-report, and a coordinator (TemporalChainExtractor) merges the reports into a single VIDEO_TOPOLOGY_BLOCK carrying an evidence_trace listing the contributing agents.

Claim 3. A cross-frame object identity method wherein a per-frame object detector emits bounding boxes without stable IDs, and an IoU-based greedy-matching algorithm assigns stable cross-frame track identities by matching consecutive-frame detections with IoU above a CANNOT_MUTATE threshold and matching label, such that downstream motion, impact, and causality agents operate on stable Track objects rather than raw per-frame detections.

Claim 4. An impact detection method operating on ObjectTracker output that fires a contact event when two tracks' bounding boxes overlap (IoU greater than contact_threshold) and at least one track had non-trivial velocity in preceding frames, and fires a deceleration event when a single track's velocity drops sharply across two consecutive frames — with both events carrying the participating track IDs, frame index, impact type, and magnitude, and the event report HMAC-signed under the video signing key.

Claim 5. A temporal causality chain method wherein a TemporalChainExtractor merges ObjectTrackReport, MotionReport, and ImpactReport into an ordered sequence of typed TemporalEvents (appear / motion_start / contact / motion_change / disappear) representing the full event structure of the clip, and the chain is serialised as a compact JSON array and HMAC-signed under a namespace key (axiom-video-temporal-v1) distinct from the object-tracking key (axiom-video-objects-v1) — providing domain isolation between the tracking and causality signing surfaces.

Claim 6. A cross-modal sensory composition method wherein a VIDEO_TOPOLOGY_BLOCK (ORVL-027) and an AUDIO_GROOVE_BLOCK (ORVL-026) are merged by a multimodal coordinator into a single signed evidence graph, with video supplying object identities, surface states, and physical causality, and audio supplying timing, rhythm, tone, and intent — and the merged graph constituting the Audio + Video layers of an Axiom Event Token (ORVL-025), verified as a unit under the Event Token signing key.

> *"Store the topology, not the thundercloud. Video Topology gives every agent the event structure — objects, motion, cause — without a single pixel riding along."*